2) Hospital Healthcare Analytics.

Data collection Source

- Reg. forms → Demography, Age, Sex, Insurance (structured)
- lab reports → Blood, pathology, culture test (tabular)
- medical devices → IoT, ECG / BP, Vitals. (Time series)
- doctor notes. → Symptoms, diagnosis (unstructured)

↓

Ingestion & processing.

↓

→ NLP / named entity Recog ...
→ signal processing.
→ Relational join.

↓

Healthcare Analytics.

↓

clinical DS ⟷ operational Analytics

clinical DS:
- early sepsis prediction
- Diagnosis recommendation

operational Analytics:
- Resource & staff allot
- Patient wait-time.

3) E-commerce Data Hygiene:-

Identified Issues.

↓

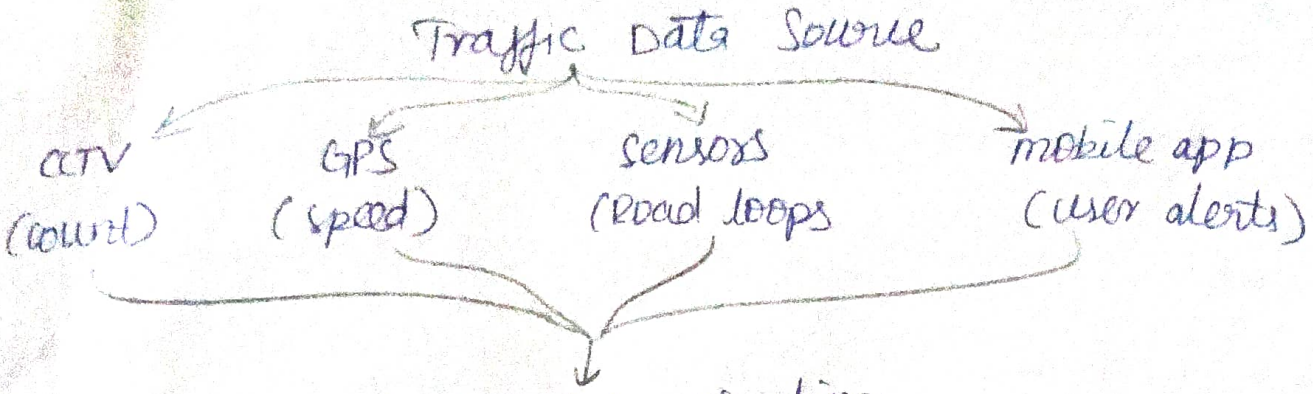
Data cleaning workflow.

- missing customer details → drop row if P key is null → impute non-critical fields
- Duplicate orders → Identify composite key → Deduplicate
- Wrong product entity → Cross check with catalog → Reject SKU format
- unusual purchase amt → outlier detection → validate high values

↓

Verified & consistent Dataset.

4) Smart city Traffic:



1. Ingestion pipelines.
↓
collect raw live data.
↓
2. Alignment & Synchronization.
-> Spatial - match to road segment IDs.
-> Temporal - sync by 5 min intervals.
↓
3. merging & conflict checks.
->Combine Speeds & vehicle counts.
-> solve overlap sensors report.
↓
Unified Traffic Dataset for Analysis.

5) Bank loan Approval-

